2. LIMITATIONS

2.1 KINDS OF OPERATION

The cargo hook system is approved for lifting external loads which are jettisonable and lifted free of land or water during rotorcraft operation.

Operations with a load attached to the suspension assembly have to be conducted in accordance with the appropriate operating rules for external loads.

External cargo operation of the helicopter is approved according to VFR.

Operation of the helicopter with no load suspended from the external cargo hook is authorized under normal airworthiness certificate without removing the hook from the helicopter.

2.2 OCCUPANTS

Only those persons necessary for the accomplishment of the work activity directly associated with external cargo operation may be carried in the helicopter.

2.3 MASS AND LOAD LIMITS

2.3.1 Max Gross Mass

No change in the basic Flight Manual data.

2.3.2 Max External Load Mass

Max external load mass . . . . 900 kg (1984 lb)

2.4 CENTER OF GRAVITY LIMITATIONS

2.4.1 Longitudinal Center of Gravity

The loading distribution of the helicopter before external load pick-up shall comply with the gross mass and center of gravity limitations for the basic helicopter (Section 2, basic FLM).

EFFECTIVITY *Variants CB, CB-2, CDN-B, CBS, CBS-2 and CDN-BS*

Any jettisonable external load up to the mass limit defined above may be attached to the cargo hook provided the maximum combined gross mass of 2400 kg (helicopter plus external load) is not exceeded (see Fig. 1).

EFFECTIVITY *Variants CB-4, CDN-B-4, CBS-4, and CDN-BS-4*

Any jettisonable external load up to the mass limit defined above may be attached to the cargo hook provided the maximum combined gross mass of 2500 kg (helicopter plus external load) is not exceeded (see Fig. 2).

EFFECTIVITY *ALL*

2.5 AIRSPEED LIMITS

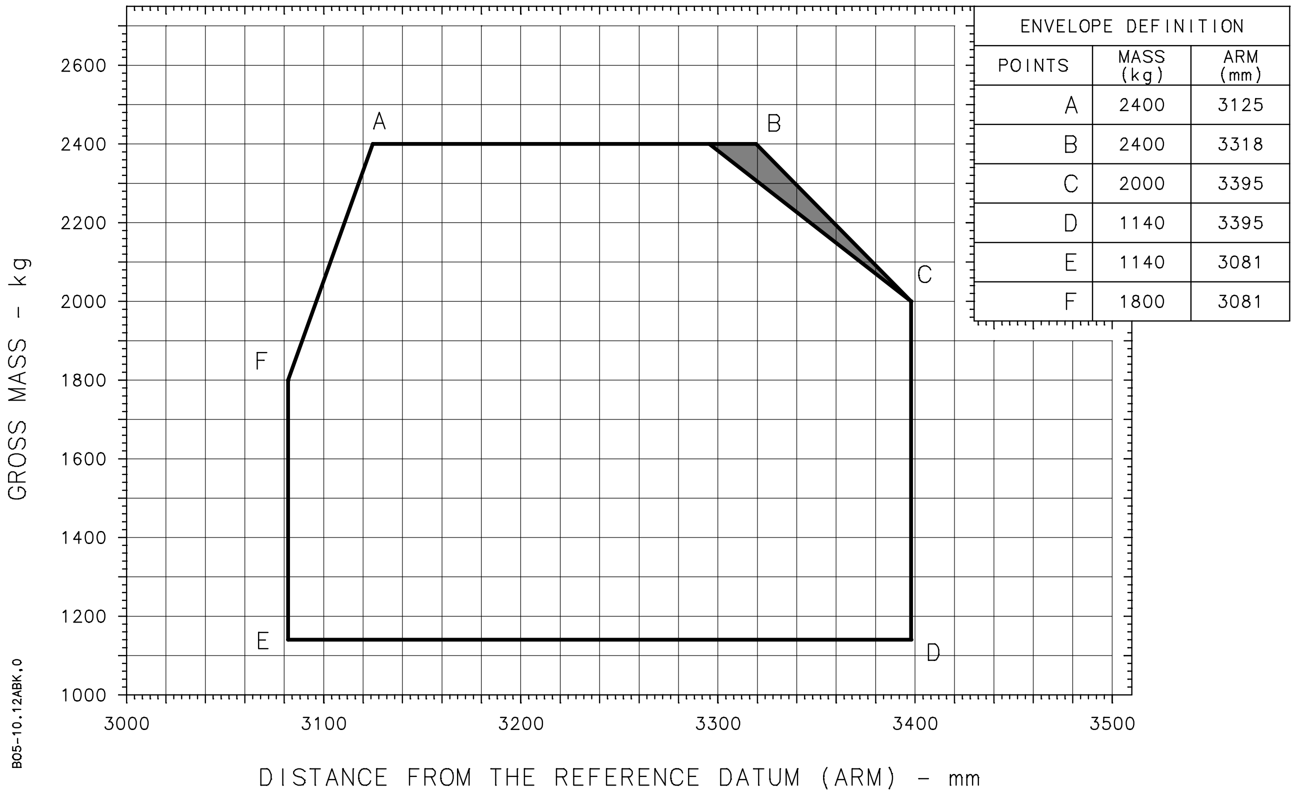
The airspeed limitations of the basic FLM, Section 2 apply when the hook assembly is secured in its stowed position or retracted.

Max airspeed with external load .... 100 KIAS

Max airspeed when trailing hook assembly (no load) and during extension or retraction of the hook assembly .... 60 KIAS

NOTE It is the pilot's responsibility to reduce airspeed and bank angle to the extent required to ensure safe operation under the prevailing circumstances.

EFFECTIVITY *Variants CB, CB-2, CDN-B, CBS, CBS-2 and CDN-BS*



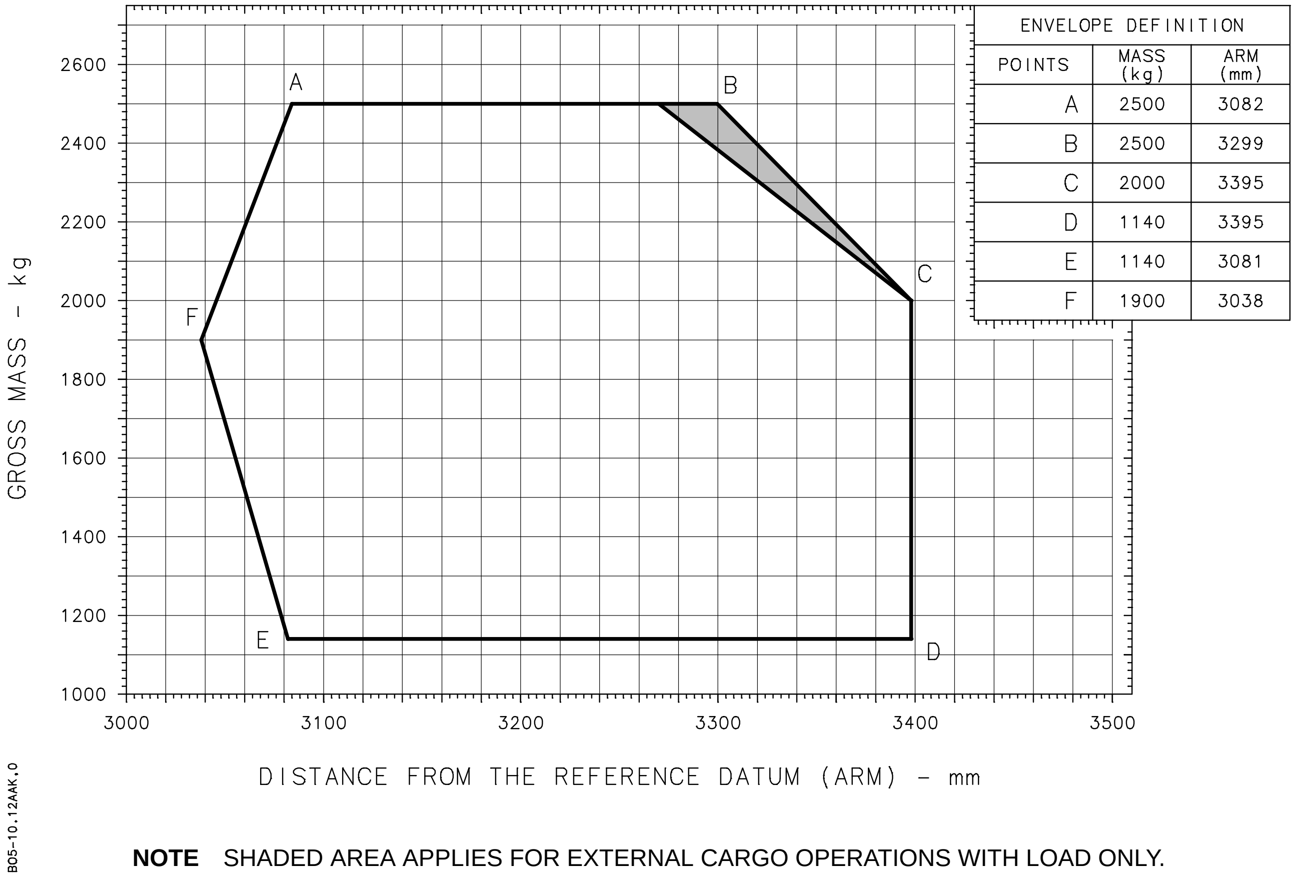
ENVELOPE DEFINITION

POINTS	MASS (kg)	ARM (mm)
A	2400	3125
B	2400	3318
C	2000	3395
D	1140	3395
E	1140	3081
F	1800	3081

NOTE SHADED AREA APPLIES FOR EXTERNAL CARGO OPERATIONS WITH LOAD ONLY.

Fig. 1 Longitudinal CG External Load Envelope

EFFECTIVITY *Variants CB-4, CDN-B-4, CBS-4, and CDN-BS-4*



ENVELOPE DEFINITION		
POINTS	MASS (kg)	ARM (mm)
A	2500	3082
B	2500	3299
C	2000	3395
D	1140	3395
E	1140	3081
F	1900	3038

NOTE SHADED AREA APPLIES FOR EXTERNAL CARGO OPERATIONS WITH LOAD ONLY.

Fig. 2 Longitudinal CG External Load Envelope

EFFECTIVITY *ALL*

2.6 EXTERNAL LOAD OPERATIONAL LIMITS

WARNING HELICOPTER HANDLING CHARACTERISTICS MAY BE AFFECTED BY THE SIZE, MASS, AND SHAPE OF THE EXTERNAL LOAD BEING CARRIED. IN PARTICULAR, LOADS OF RELATIVELY LOW MASS GENERATING SIGNIFICANT AERODYNAMIC REACTIONS ON ACCOUNT OF THEIR SIZE AND/OR SHAPE MAY BECOME UNSTABLE.

ANY UNSTABLE LOAD MAY JUMP, OSCILLATE, OR ROTATE RESULTING IN LOSS OF CONTROL, UNDUE STRESS ON AND/OR CONTACT WITH THE HELICOPTER CAUSING POSSIBLE CATASTROPHIC RESULTS.

THEREFORE, EACH OPERATOR SHALL ESTABLISH ADEQUATE OPERATIONAL LIMITS AND PROCEDURES THAT PRECLUDE THE POSSIBILITY OF THE LOAD CONTACTING THE HELICOPTER.

THE PILOT, THROUGH USE OF AN EXTERNALLY MOUNTED REAR-VIEW MIRROR, OR A TRAINED ON-BOARD OBSERVER SHALL MONITOR THE LOAD REACTION DURING FLIGHT IN ORDER TO REGAIN CONTROL SHOULD THE LOAD BECOME UNSTABLE.

Flight with an empty net or unballasted sling as an external load is prohibited unless approved operational limits and procedures provided by the operator allow for such an operation.

Towing loads touching the ground or water surface have not been flight demonstrated.

Immersion of a cable / rope attached to the cargo hook into water for the purpose of picking up loads is permissible within the hover speed range if obstruction clearance is fully ensured.

The distance between load and hook will depend on the type of load or operation; however, it shall be kept as short as possible.

2.7 LOAD ATTACHMENT RING OR SHACKLE DIMENSIONS

WARNING THE USE OF A LOAD ATTACHMENT RING OR SHACKLE WITH INCORRECT DIMENSIONS MAY LEAD TO LOSS OR JAMMING OF THE LOAD.

The operator is responsible for selecting an appropriate load attachment ring or shackle. The attachment means must be capable of safely carrying the load and unable to jam, to roll out or to turn out (refer to Para 8.2 of this FMS for respective information).

EFFECTIVITY *Before ASB BO105–80A–146*

The load shall only be attached to hook using a fixed ring with inner diameter between 50 and 80 mm (see placard).

EFFECTIVITY *After ASB BO105–80A–146*

The load shall only be attached to the hook using:

– a ring with dimensions as defined in Fig. 3, or

– a shackle with dimensions as defined in Fig. 4.

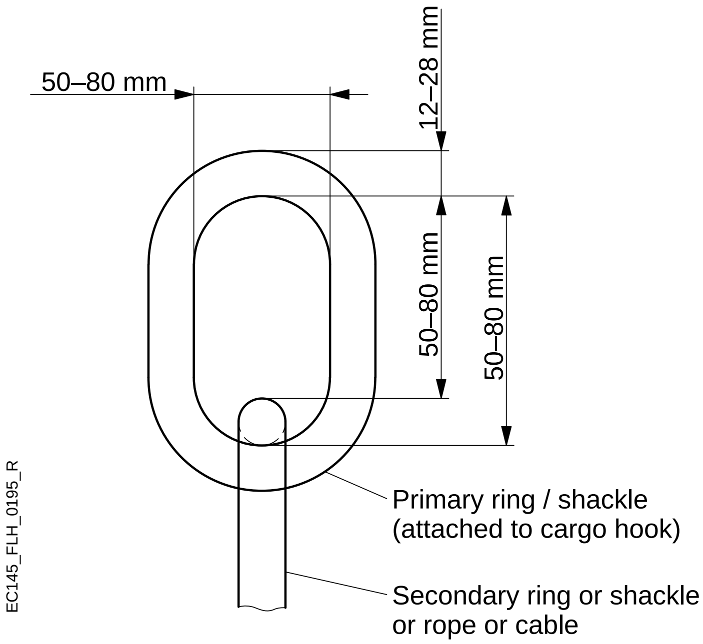


Fig.3 Required ring dimensions

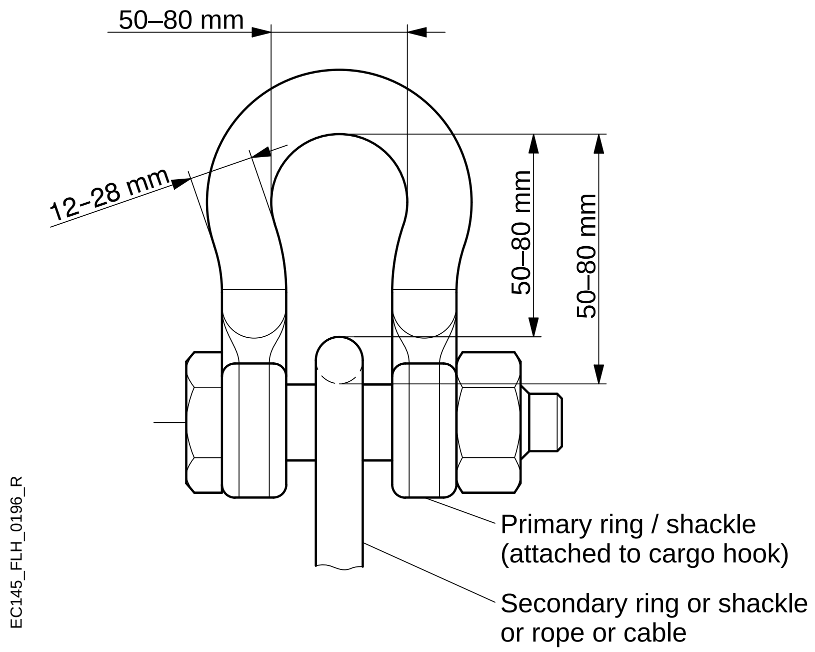


Fig.4 Required shackle dimensions

EFFECTIVITY *All*

2.8 OTHER LIMITATIONS

CAUTION RUNNING LANDINGS WITH THE HOOK DEPLOYED SHALL BE AVOIDED.

2.9 PLACARDS AND DECALS

Text:

EXTERNAL LOAD HOOK	
ONLY PERSONS DIRECTLY NECESSARY FOR THE WORK ACTIVITY MAY BE CARRIED IN THE HELICOPTER	
MAX AIRSPEED:	
WITH LOAD	**100 KTS**
WITHOUT LOAD (HOOK UNSTOWED)	**60 KTS**

Location: Upper RH windshield frame, in pilot's field-of-view.

Text: (only if hook winch is not installed)

Hier nur Lasthaken
ohne Last einhängen.

Install load hook
with no load only.

Location: LH fuselage adjacent hook retainer.

EFFECTIVITY *All Hook Assemblies Except P/N 117-80127*

Text:

MAX. AUSSENLAST 900kg
MAX. EXTERNAL LOAD 900kg (1984lb.)

Location: Hook assembly housing, RH side.

EFFECTIVITY *Hook Assembly P/N 117-80127*

Text:

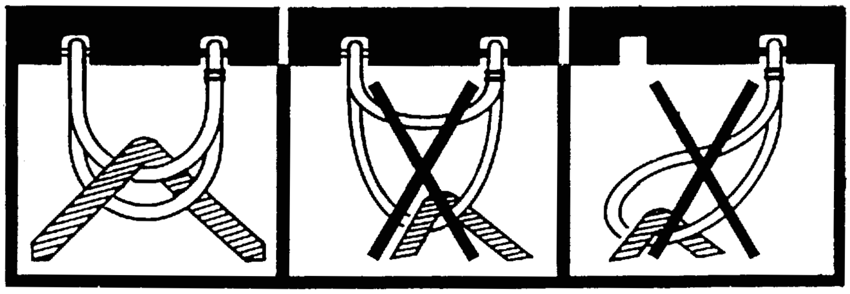
MAX. AUSSENLAST SIEHE FLUGHANDBUCH **MAX. EXTERNAL LOAD SEE FLIGHT MANUAL**

Location: Hook assembly housing, RH side.

EFFECTIVITY *Hook Assembly P/N 105-80332*

Text:

BO5-10.12AAM,0

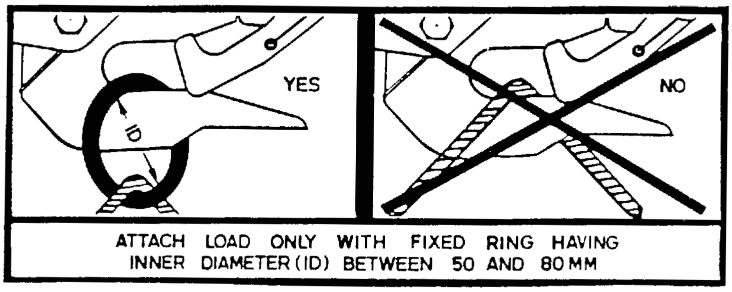


Location: Hook assembly mechanical release cover, both sides.

EFFECTIVITY *Hook Assemblies P/N 105-80337 and 117-80127 and before ASB BO105–80A–146*

Text:

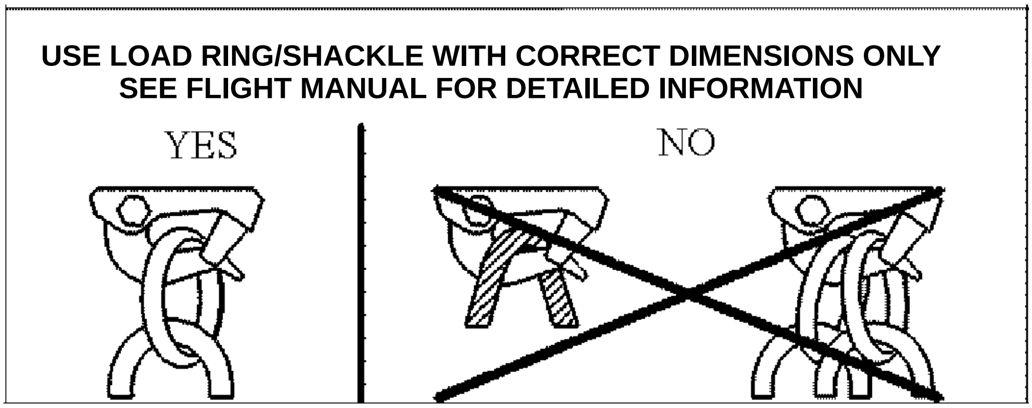
BO5−10.12 ACM,0



Location: Hook assembly housing,

EFFECTIVITY *Hook Assembly P/N 105-80337 and 117-80127 and after ASB BO105–80A–146*

Text:



Location: Hook assembly housing, both sides

EFFECTIVITY *ALL*